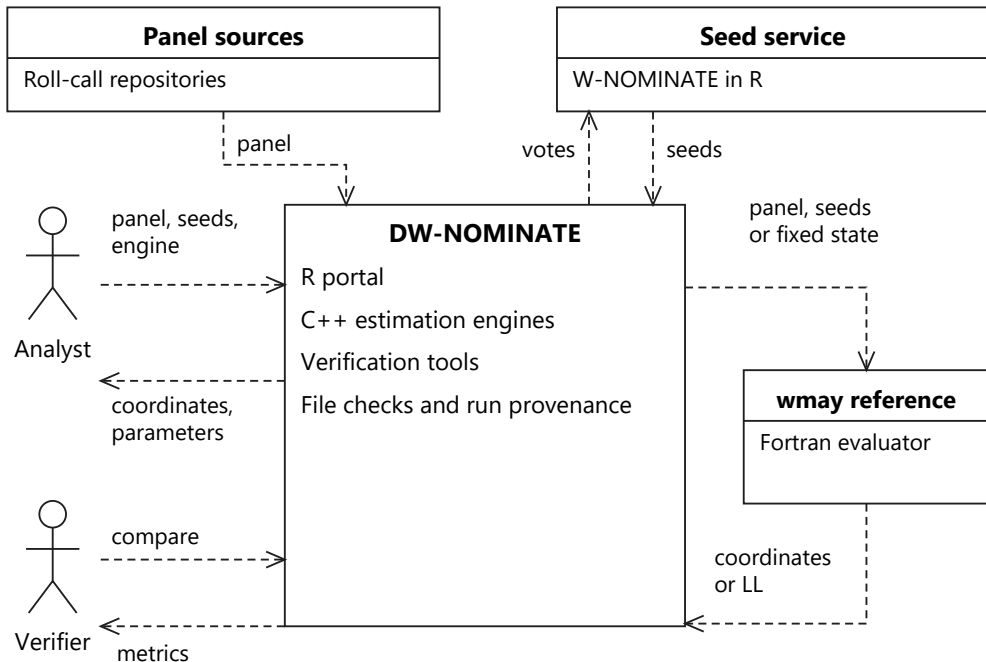
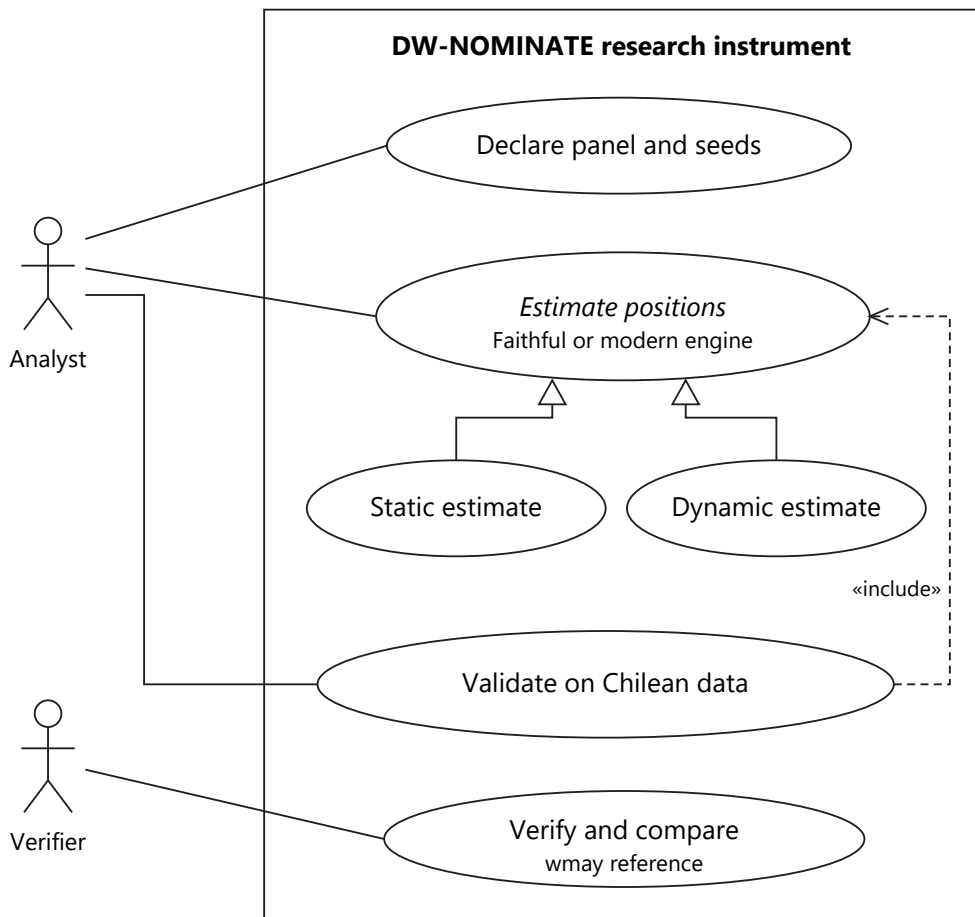


Instrument context



Research use cases



Legacy responsibility mapping

Fortran predecessors

Input and execution

2004: file-based program
wmay: dwnom + R/CLI wrapper

Configuration and state

2004: COMMON; wmay: modules
Shared parameters and arrays

Parameter searches

WINT, SIGMAS, RCINT2, XINT
GRID safeguard: 2004 only

Cutting geometry

CUTPLANE, SEARCH, JAN11PT
2004: LSVRR; wmay: DGESDD

Likelihood

PLOG; PROLLC2, PROX

Numerical support

RSORT, REGA, normal table

Faithful C++

Input and orchestration

CLI and CSVLoader
Cycle in DWNominate

Configuration and state

DWNominateConfig
State held by DWNominate

Translated searches

Scalar, roll-call, legislator
GRID selected by lineage

Cutting geometry

cutting_plane, cutting_point
DGESDD in the matched profile

Likelihood and derivatives

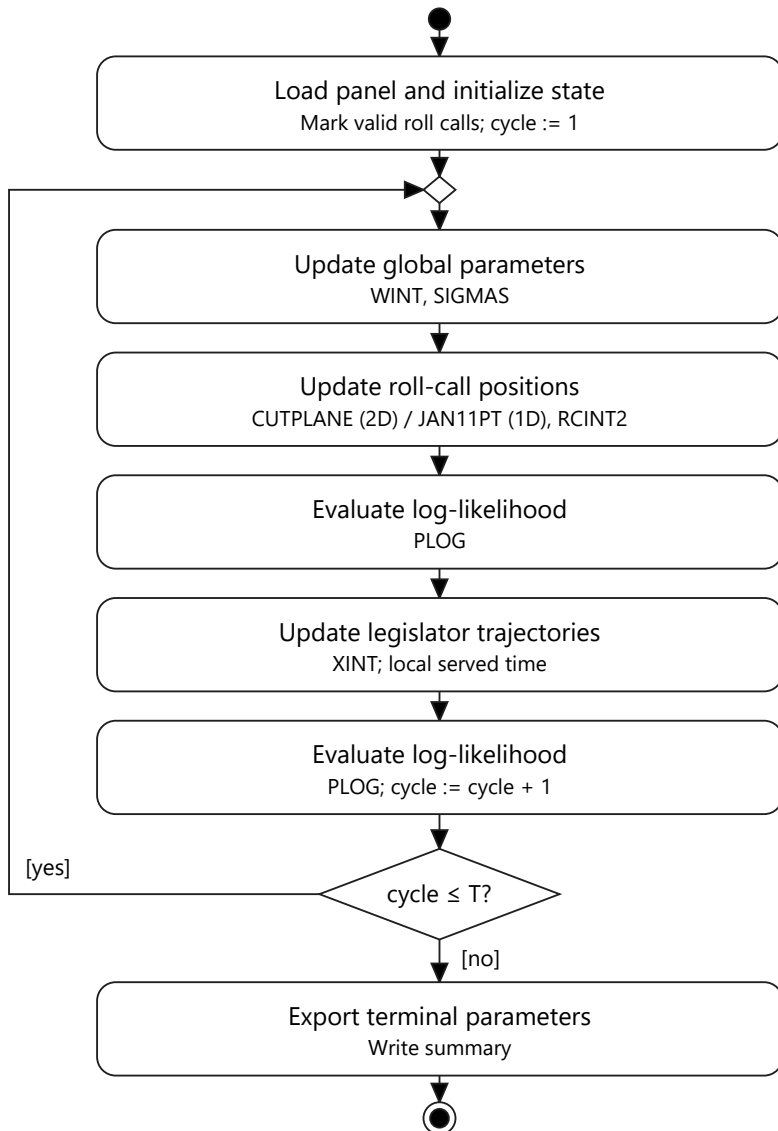
likelihood
Derivative modules

Numerical support

sort_utils, simple_ols
normal_cdf

Each horizontal link maps a responsibility, not a dependency.

Fixed-cycle estimation



Scope: faithful in wmay mode; 2D, $T = 4$. No convergence test.

Known limitation: faithful summary mixes terminal LL and earlier classification.

Optimization mechanisms

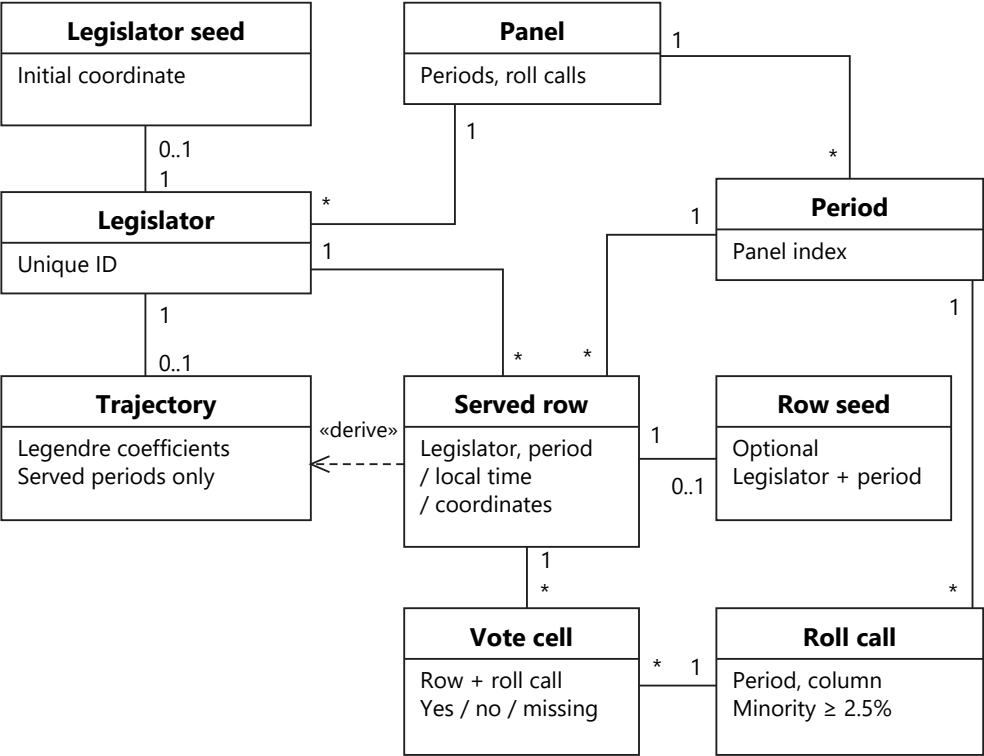
Controlled profile: local scalar searches, strict arithmetic, COBYLA blocks.

Parameter	Faithful	Modern	Constraint
2nd-dimension weight	WINT grid	BOBYQA scalar	Local WINT interval; 2D only
Beta	SIGMAS grid	BOBYQA scalar	Local SIGMAS interval
Roll call	RCINT2 term-wise search	COBYLA block	Midpoint norm ≤ 1
Legislator	XINT term-wise search	COBYLA block	Constant-term norm ≤ 1

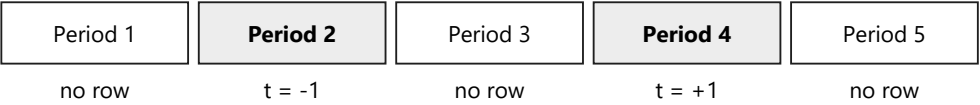
Global scalar search, SLSQP and hybrid blocks are outside this profile.

Replication removes global scalar bounds; the local search interval remains.

Data and served time

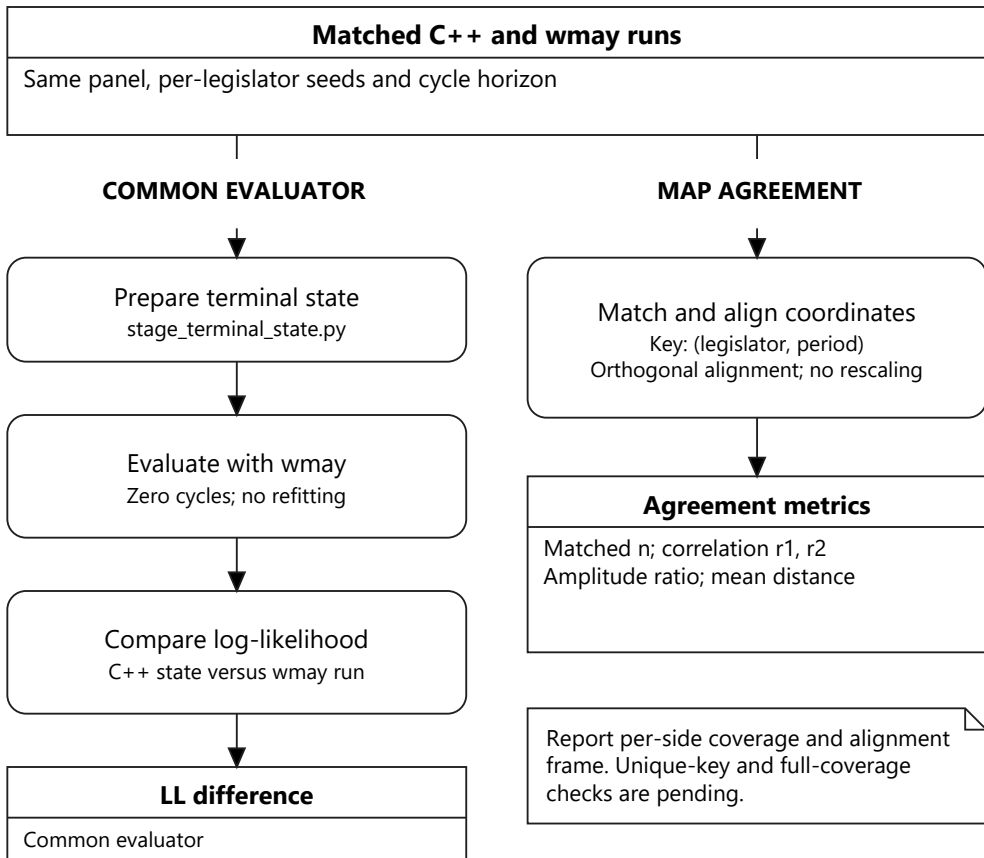


Example: one legislator serves periods 2 and 4



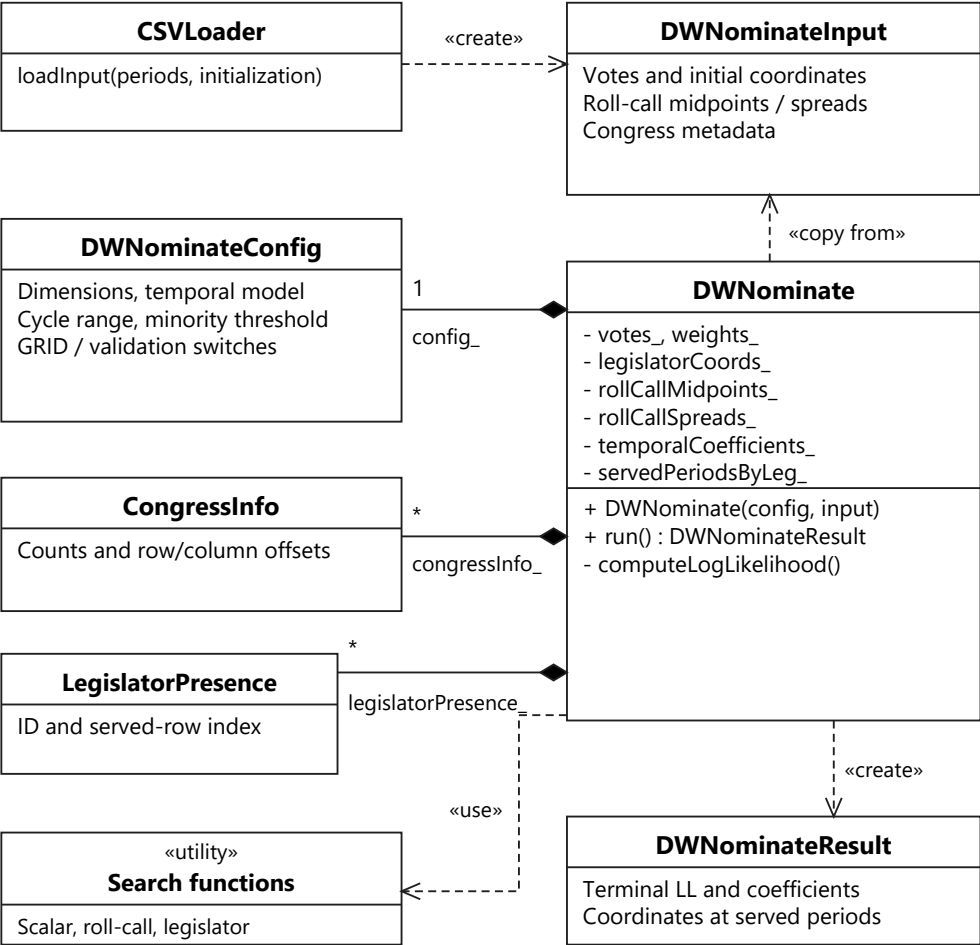
Two served periods $\rightarrow$ constant trajectory. Linear / quadratic / cubic need 5 / 6 / 7.

Two complementary checks



Reference here: wmay. This protocol does not establish original-2004 equivalence.

Faithful engine: state ownership



Filled diamond: ownership by value. Selected members; input is not retained.

One estimation request

